

SF1811 Optimization

Lecture 11: Unconstrained nonlinear optimization and Newton's method

Chapters 13–14 and 16

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Unconstrained nonlinear optimization and Newton's method

Main question:

How can a nonlinear function be minimized by repeatedly solving quadratic models?

Today:

- Local optimality conditions.
- Descent directions.
- Local quadratic models.
- Newton's direction and Newton's method.
- Convergence behavior and possible difficulties.

Unconstrained nonlinear optimization

Consider

$$(P) \qquad \underset{x \in \mathbb{R}^n}{\text{minimize}}\, f(x),$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable.

A point $\hat{x}$ is a local minimizer if there is an $\epsilon > 0$ such that

$$f(x) \geq f(\hat{x})$$

whenever

$$\|x - \hat{x}\| < \epsilon.$$

A point $\hat{x}$ is a global minimizer if

$$f(x) \geq f(\hat{x}) \qquad \text{for every } x \in \mathbb{R}^n.$$

For a general nonlinear function, a local minimizer need not be global.

Restricting the function to a line

Fix a point $x \in \mathbb{R}^n$ and a direction $d \in \mathbb{R}^n$, and define

$$\varphi(t) = f(x + td), \qquad t \in \mathbb{R}.$$

By the chain rule,

$$\varphi'(t) = \nabla f(x + td)d,$$

and

$$\varphi''(t) = d^{\mathsf{T}} F(x + td)d,$$

where $F(x)$ is the Hessian of f at x.

In particular,

$$\varphi'(0) = \nabla f(x)d, \qquad \varphi''(0) = d^{\mathsf{T}} F(x)d.$$

Multivariable optimality can be studied one direction at a time.

First-order necessary condition

Theorem

If $\hat{x}$ is a local minimizer of f, then $\nabla f(\hat{x}) = 0$.

Proof

For an arbitrary direction d, define

$$\varphi(t) = f(\hat{x} + td).$$

Since $\hat{x}$ is a local minimizer of f, the point $t = 0$ is a local minimizer of φ. Hence

$$0 = \varphi'(0) = \nabla f(\hat{x})d.$$

This holds for every $d \in \mathbb{R}^n$, and therefore

$$\nabla f(\hat{x}) = 0.$$

Second-order necessary condition

Theorem

If $\hat{x}$ is a local minimizer of f, then

$$\nabla f(\hat{x}) = 0 \qquad \textit{and} \qquad F(\hat{x}) \succeq 0.$$

Proof

We already know that $\nabla f(\hat{x}) = 0$. For an arbitrary direction d, let

$$\varphi(t) = f(\hat{x} + td).$$

Since $t = 0$ is a local minimizer of φ, $\varphi''(0) \geq 0$. But

$$\varphi''(0) = d^{\mathsf{T}} F(\hat{x}) d.$$

Thus $d^{\mathsf{T}} F(\hat{x}) d \geq 0$ for every d, which means that $F(\hat{x}) \succeq 0$.

Second-order sufficient condition

Theorem

If

$$\nabla f(\hat{x}) = 0 \qquad \textit{and} \qquad F(\hat{x}) \succ 0,$$

then $\hat{x}$ is a strict local minimizer of f.

Since F is continuous and $F(\hat{x}) \succ 0$, there is a neighborhood of $\hat{x}$ in which

$$d^{\mathsf{T}} F(x) d > 0 \qquad \text{for every } d \neq 0.$$

Taylor's formula gives, for $x = \hat{x} + d$,

$$f(\hat{x} + d) = f(\hat{x}) + \nabla f(\hat{x}) d + \frac{1}{2} d^{\mathsf{T}} F(\hat{x} + \theta d) d$$

for some $\theta \in (0, 1)$.

For nonzero d sufficiently small,

$$f(\hat{x} + d) > f(\hat{x}).$$

Necessary and sufficient conditions

At a local minimizer, necessarily

$$\nabla f(\hat{x}) = 0, \qquad F(\hat{x}) \succeq 0.$$

A sufficient condition for a strict local minimum is

$$\nabla f(\hat{x}) = 0, \qquad F(\hat{x}) \succ 0.$$

The gap between the conditions is real:

- For $f(x) = x^3$ at $\hat{x} = 0$,

  $$f'(0) = 0, \qquad f''(0) = 0,$$

  but 0 is not a local minimizer.
- For $f(x) = x^4$ at $\hat{x} = 0$,

  $$f'(0) = 0, \qquad f''(0) = 0,$$

  and 0 is a strict global minimizer.

A semidefinite Hessian at a stationary point is inconclusive.

Descent directions

A direction d is a descent direction for f at x if

$$f(x + td) < f(x)$$

for every positive t sufficiently close to zero.

If

$$\nabla f(x)d < 0,$$

then d is a descent direction.

Indeed,

$$f(x + td) = f(x) + t\nabla f(x)d + o(t),$$

and the negative linear term dominates the remainder for positive t sufficiently close to zero.

If $\nabla f(x) \neq 0$, the choice $d = -\nabla f(x)^{\mathsf{T}}$ gives $\nabla f(x)d = -\|\nabla f(x)\|^2 < 0$.

The negative gradient is always a descent direction away from a stationary point.

Why use second-order information?

The negative gradient uses only first-order information:

$$d = -\nabla f(x)^{\mathsf{T}}.$$

However, the objective may have very different curvature in different directions.

The Hessian scales and combines directions according to local curvature.

A local quadratic model

At the current iterate $x^{(k)}$, approximate $f(x^{(k)} + d)$ by

$$q_k(d) = f(x^{(k)}) + \nabla f(x^{(k)})d + \frac{1}{2}d^{\mathsf{T}}F(x^{(k)})d.$$

This is the second-order Taylor model of f at $x^{(k)}$.

If

$$F(x^{(k)}) \succ 0,$$

then q_k is a strictly convex quadratic function of d and has a unique minimizer.

Replace the nonlinear objective locally by a quadratic optimization problem.

Newton's direction

The gradient of the quadratic model is

$$\nabla q_k(d) = \nabla f(x^{(k)}) + d^{\mathsf{T}} F(x^{(k)}).$$

Setting this gradient to zero gives

$$F(x^{(k)})d^{(k)} = -\nabla f(x^{(k)})^{\mathsf{T}}.$$

The solution $d^{(k)}$ is the Newton direction.
If $F(x^{(k)}) \succ 0$, then

$$\nabla f(x^{(k)})d^{(k)} = -{d^{(k)}}^{\mathsf{T}} F(x^{(k)})d^{(k)} < 0$$

whenever $d^{(k)} \neq 0$.
Thus the Newton direction is then a descent direction.

Newton's method

Given an initial point $x^{(0)}$, repeat:

1. Solve the Newton equation
$$F(x^{(k)})d^{(k)} = -\nabla f(x^{(k)})^{\mathsf{T}}.$$
2. Choose a step length $t_k > 0$.
3. Update
$$x^{(k+1)} = x^{(k)} + t_k d^{(k)}.$$

The choice
$$t_k = 1$$
is called a full Newton step.

Near a solution satisfying the second-order sufficient condition, full Newton steps are typically appropriate. Farther away, a shorter step may be needed.

Each Newton iteration solves one linear system.

Newton as linearization of the gradient

A minimizer is expected to satisfy

$$\nabla f(x) = 0.$$

Linearize the transposed gradient at $x^{(k)}$:

$$\nabla f(x^{(k)} + d)^{\mathsf{T}} \approx \nabla f(x^{(k)})^{\mathsf{T}} + F(x^{(k)})d.$$

Set the linear approximation equal to zero:

$$\nabla f(x^{(k)})^{\mathsf{T}} + F(x^{(k)})d = 0.$$

This gives the Newton equation

$$F(x^{(k)})d = -\nabla f(x^{(k)})^{\mathsf{T}}.$$

Newton's direction therefore has two equivalent interpretations:

- minimize a local quadratic model of f;
- apply Newton's root-finding method to $\nabla f(x) = 0$.

Quadratic functions are solved in one iteration

Let

$$f(x) = \frac{1}{2}x^{\mathsf{T}}Hx + c^{\mathsf{T}}x + c_0, \qquad H \succ 0.$$

Then

$$\nabla f(x)^{\mathsf{T}} = Hx + c, \qquad F(x) = H.$$

The Newton equation is

$$Hd = -(Hx + c).$$

Therefore,

$$H(x + d) = -c,$$

and a full Newton step gives

$$x^{(1)} = x + d = -H^{-1}c.$$

This is the unique minimizer, independently of the starting point.

Newton's method is exact for strictly convex quadratic functions.

A nonlinear example

Consider the one-variable function

$$f(x) = \frac{1}{4}x^4 - \frac{1}{2}x^2 + x.$$

Then

$$f'(x) = x^3 - x + 1, \qquad f''(x) = 3x^2 - 1.$$

Newton's iteration for finding a stationary point is

$$x^{(k+1)} = x^{(k)} - \frac{(x^{(k)})^3 - x^{(k)} + 1}{3(x^{(k)})^2 - 1}.$$

Unlike the quadratic case:

- the Hessian changes between iterations;
- more than one iteration is generally needed;
- a poor starting point may give undesirable behavior.

When can Newton's method fail?

The basic Newton method is a local method. Difficulties may occur when:

- $F(x^{(k)})$ is singular, so the Newton equation may have no unique solution;
- $F(x^{(k)})$ is indefinite, so the Newton direction need not be a descent direction;
- a full Newton step increases the objective value;
- the starting point is too far from a desired minimizer;
- the method approaches a stationary point that is not a minimizer.

Common safeguards include:

- choosing a shorter step length;
- modifying the Hessian to obtain positive definiteness;
- combining Newton's method with a more robust descent method.

Newton's method is fast near a suitable solution, but needs safeguards to be globally reliable.

Local convergence of Newton's method

Suppose that $\hat{x}$ satisfies

$$\nabla f(\hat{x}) = 0, \qquad F(\hat{x}) \succ 0,$$

and that the Hessian is sufficiently regular near $\hat{x}$.

If the starting point is sufficiently close to $\hat{x}$, full Newton steps converge rapidly to $\hat{x}$.

The characteristic local behavior is quadratic convergence:

$$\|x^{(k+1)} - \hat{x}\| \leq C\|x^{(k)} - \hat{x}\|^2$$

for all sufficiently large k and some constant $C > 0$.

Thus, once the iterates are close enough, the number of correct digits may approximately double in each iteration.

Main ideas

After this lecture, you should be able to:

- state the first- and second-order necessary conditions for a local minimizer;
- state a second-order sufficient condition for a strict local minimizer;
- identify a descent direction using the gradient;
- construct the local quadratic Taylor model;
- derive Newton's direction by minimizing the model;
- interpret Newton's method as root finding for $\nabla f(x) = 0$;
- perform Newton iterations in small examples;
- explain why a strictly convex quadratic function is minimized in one full Newton step;
- describe situations in which safeguards are needed.

Newton's method minimizes a sequence of local quadratic models.